Worksheet 2.4 — Polynomial Inequalities

Grade 12 Advanced Functions (MHF4U) · Unit 2: Polynomial Equations and Inequalities

The Big Ideas

Find the zeros, then use a **sign table** to find the sign of the polynomial on every interval. The sign flips at odd multiplicity but not at even multiplicity.

- Move everything to one side, factor completely, and find the zeros (with their multiplicities).
- Build a sign table: a column for each interval and each zero, a row for each factor and one for the product.
- Multiply the signs in each column ($- \times - = +$); keep the columns that satisfy the inequality, and include the zeros only for $\geq$ or $\leq$.

Core Concepts

Beyond equations

A polynomial **inequality** asks where the graph lies above or below the axis, not just where it crosses. Always move everything to one side first so the other side is 0; never divide by a factor that contains x.

Use the zeros

Factor completely and set each factor to zero. The zeros split the number line into intervals; the polynomial keeps one sign on each, so one test point per interval is enough.

Build a sign table

Make a column for every interval *and* every zero, a row for every factor, and a row for the product. Fill in the sign of each factor, then multiply the signs down each column: an even number of negatives gives $+$, an odd number gives $-$. The table below solves $(x-1)(x+2) > 0$; the ✓ columns satisfy it.

Interval	$x < -2$	$x = -2$	$-2 < x < 1$	$x = 1$	$x > 1$
Test point	-3	–	-1	–	2
$x - 1$	$-$	$-$	$-$	0	$+$
$x + 2$	$-$	0	$+$	$+$	$+$
$P(x)$	$+$	0	$-$	0	$+$
$P(x) > 0$?	✓	×	×	×	✓

Reading the sign

Keep the columns that satisfy the inequality. Include the zero columns only for $\geq$ or $\leq$. An odd-multiplicity zero changes the sign; an even-multiplicity zero does not (the graph touches the axis and turns back). For a positive leading coefficient, the right-most interval is always $+$.

Worked Examples

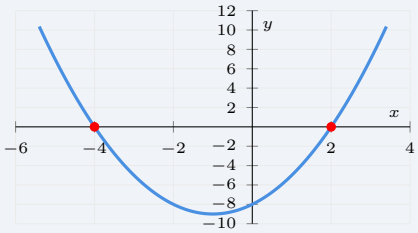
Example 1 — Product positive

Solve $(x-2)(x+4) > 0$.

Solution. **Step 1:** It is already compared with 0 and factored.
Step 2: Zeros: $x-2=0 \Rightarrow x=2$ and $x+4=0 \Rightarrow x=-4$, each of multiplicity 1, so the sign changes at each. They split the number line into $x<-4$, $-4<x<2$ and $x>2$.
Step 3: Sign table (one test point in each interval):

Interval	$x<-4$	$x=-4$	$-4<x<2$	$x=2$	$x>2$
Test point	-5	–	-3	–	3
$x-2$	$-$	$-$	$-$	0	$+$
$x+4$	$-$	0	$+$	$+$	$+$
$P(x)$	$+$	0	$-$	0	$+$
$P(x)>0$?	✓	×	×	×	✓

Step 4: Keep the ✓ columns ($P>0$). The zeros give 0, which is not >0, so they are excluded.
Answer: $x<-4$ or $x>2$, i.e. $(-\infty,-4)\cup(2,\infty)$. The graph is above the axis there:



Example 2 — Product negative

Solve $(x-2)(x+4) < 0$.

Solution. **Step 1:** Same factors and zeros as Example 1, so the same table. Now we want $P<0$.

Interval	$x<-4$	$x=-4$	$-4<x<2$	$x=2$	$x>2$
Test point	-5	–	-3	–	3
$x-2$	$-$	$-$	$-$	0	$+$
$x+4$	$-$	0	$+$	$+$	$+$
$P(x)$	$+$	0	$-$	0	$+$
$P(x)<0$?	×	×	✓	×	×

Step 2: Keep the $-$ column. The strict inequality excludes the zeros.
Answer: $-4<x<2$, i.e. $(-4,2)$.

Example 3 — Factor first

Solve $x^2 + x - 12 > 0$.

Solution. **Step 1:** Factor: two numbers with product -12 and sum 1 are 4 and -3, so $(x+4)(x-3) > 0$.
Step 2: Zeros -4 and 3; sign table:

Interval	$x < -4$	$x = -4$	$-4 < x < 3$	$x = 3$	$x > 3$
Test point	-5	–	-3	–	4
$x + 4$	$-$	0	$+$	$+$	$+$
$x - 3$	$-$	$-$	$-$	0	$+$
$P(x)$	$+$	0	$-$	0	$+$
$P(x) > 0$?	✓	×	×	×	✓

Step 3: Keep the $+$ columns.
Answer: $x < -4$ or $x > 3$, i.e. $(-\infty, -4) \cup (3, \infty)$.

Example 4 — Three factors

Solve $x(x + 4)(x - 1) > 0$.

Solution. **Step 1:** Zeros -4, 0, 1, all of multiplicity 1: four intervals, and the sign of P alternates from one to the next.

Interval	$x < -4$	$x = -4$	$-4 < x < 0$	$x = 0$	$0 < x < 1$	$x = 1$	$x > 1$
Test point	-5	–	-3	–	0.5	–	2
x	$-$	$-$	$-$	0	$+$	$+$	$+$
$x + 4$	$-$	0	$+$	$+$	$+$	$+$	$+$
$x - 1$	$-$	$-$	$-$	$-$	$-$	0	$+$
$P(x)$	$-$	0	$+$	0	$-$	0	$+$
$P(x) > 0$?	×	×	✓	×	×	×	✓

Step 2: Keep the $+$ columns. Check: for a positive leading coefficient the right-most interval is $+$.
Answer: $-4 < x < 0$ or $x > 1$, i.e. $(-4, 0) \cup (1, \infty)$.

Example 5 — Even multiplicity

Solve $(x - 3)^2(x + 2) > 0$.

Solution. **Step 1:** Zeros: $x = 3$ (multiplicity 2, even) and $x = -2$ (multiplicity 1). The factor $(x - 3)^2$ is never negative, so it does not change the sign of P at $x = 3$.

Interval	$x < -2$	$x = -2$	$-2 < x < 3$	$x = 3$	$x > 3$
Test point	-3	–	-1	–	4
$(x - 3)^2$	$+$	$+$	$+$	0	$+$
$x + 2$	$-$	0	$+$	$+$	$+$
$P(x)$	$-$	0	$+$	0	$+$
$P(x) > 0$?	×	×	✓	×	✓

Step 2: P is positive on both sides of $x = 3$, but $P(3) = 0$ is not > 0, so $x = 3$ is left out.
Answer: $x > -2$, $x \neq 3$, i.e. $(-2, 3) \cup (3, \infty)$.

Example 6 — Difference of squares

Solve $x^2 - 16 < 0$.

Solution. **Step 1:** Factor: $x^2 - 16 = (x-4)(x+4)$, so the zeros are 4 and -4.

Interval	$x < -4$	$x = -4$	$-4 < x < 4$	$x = 4$	$x > 4$
Test point	-5	–	-3	–	5
$x - 4$	$-$	$-$	$-$	0	$+$
$x + 4$	$-$	0	$+$	$+$	$+$
$P(x)$	$+$	0	$-$	0	$+$
$P(x) < 0$?	$\times$	$\times$	✓	$\times$	$\times$

Step 2: Keep the $-$ column (strict, so the zeros are excluded).
Answer: $-4 < x < 4$, i.e. $(-4, 4)$.

Example 7 — With equality

Solve $x^2 - 3x - 10 \geq 0$.

Solution. **Step 1:** Factor: two numbers with product -10 and sum -3 are -5 and 2, so $(x-5)(x+2) \geq 0$.

Interval	$x < -2$	$x = -2$	$-2 < x < 5$	$x = 5$	$x > 5$
Test point	-3	–	-1	–	6
$x - 5$	$-$	$-$	$-$	0	$+$
$x + 2$	$-$	0	$+$	$+$	$+$
$P(x)$	$+$	0	$-$	0	$+$
$P(x) \geq 0$?	✓	✓	$\times$	✓	✓

Step 2: The inequality is $\geq$, so the zero columns *are* included along with the $+$ intervals.
Answer: $x \leq -2$ or $x \geq 5$, i.e. $(-\infty, -2] \cup [5, \infty)$.

Example 8 — Three factors

Solve $(x+3)(x-1)(x-4) < 0$.

Solution. **Step 1:** Zeros -3, 1, 4, each of multiplicity 1.

Interval	$x < -3$	$x = -3$	$-3 < x < 1$	$x = 1$	$1 < x < 4$	$x = 4$	$x > 4$
Test point	-4	–	-2	–	2	–	5
$x + 3$	$-$	0	$+$	$+$	$+$	$+$	$+$
$x - 1$	$-$	$-$	$-$	0	$+$	$+$	$+$
$x - 4$	$-$	$-$	$-$	$-$	$-$	0	$+$
$P(x)$	$-$	0	$+$	0	$-$	0	$+$
$P(x) < 0$?	✓	$\times$	$\times$	$\times$	✓	$\times$	$\times$

Step 2: Keep the $-$ columns.
Answer: $x < -3$ or $1 < x < 4$, i.e. $(-\infty, -3) \cup (1, 4)$.

Example 9 — Cubic

Solve $x^3 - x < 0$.

Solution. **Step 1:** Factor: take out x, then a difference of squares: $x^3 - x = x(x-1)(x+1)$. Zeros -1, 0, 1.

Interval	$x < -1$	$x = -1$	$-1 < x < 0$	$x = 0$	$0 < x < 1$	$x = 1$	$x > 1$
Test point	-2	–	-0.5	–	0.5	–	2
x	$-$	$-$	$-$	0	$+$	$+$	$+$
$x-1$	$-$	$-$	$-$	$-$	$-$	0	$+$
$x+1$	$-$	0	$+$	$+$	$+$	$+$	$+$
$P(x)$	$-$	0	$+$	0	$-$	0	$+$
$P(x) < 0$?	✓	×	×	×	✓	×	×

Step 2: Keep the $-$ columns.
Answer: $x < -1$ or $0 < x < 1$, i.e. $(-\infty, -1) \cup (0, 1)$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Solve $(x+5)(x-3) > 0$.

Question 2

Solve $(x-4)(x+3) < 0$.

Question 3

Solve $x^2 - 36 < 0$.

Question 4

Solve $x(x-4)(x+1) > 0$.

Question 5

Solve $x^2 - 4x - 5 \geq 0$.

Question 6

Solve $x^2 - 9 > 0$.

Question 7

Solve $(x+1)(x-4) < 0$.

Question 8

Solve $x^3 - 4x < 0$.

Question 9

Solve $(x-2)^2(x+1) > 0$.

Question 10

Solve $x^2 - 5x + 6 \leq 0$.

Question 11

Solve $x(x-3)(x+2) < 0$.

Question 12

Solve $x^2 + x - 6 \geq 0$.

Question 13 — Challenge

Solve $x(x-1)(x-2)(x+1) > 0$ using a sign chart.

Question Recap & Answer Key

Question Recap

1. Solve $(x+5)(x-3) > 0$.
2. Solve $(x-4)(x+3) < 0$.
3. Solve $x^2 - 36 < 0$.
4. Solve $x(x-4)(x+1) > 0$.
5. Solve $x^2 - 4x - 5 \geq 0$.
6. Solve $x^2 - 9 > 0$.
7. Solve $(x+1)(x-4) < 0$.
8. Solve $x^3 - 4x < 0$.
9. Solve $(x-2)^2(x+1) > 0$.
10. Solve $x^2 - 5x + 6 \leq 0$.
11. Solve $x(x-3)(x+2) < 0$.
12. Solve $x^2 + x - 6 \geq 0$.
13. Solve $x(x-1)(x-2)(x+1) > 0$ using a sign chart.

✓ Answer Key

1. Zeros -5, 3; signs $+, -, +$; $x < -5$ or $x > 3$, i.e. $(-\infty, -5) \cup (3, \infty)$
2. Zeros -3, 4; signs $+, -, +$; $-3 < x < 4$, i.e. $(-3, 4)$
3. Zeros -6, 6; signs $+, -, +$; $-6 < x < 6$, i.e. $(-6, 6)$
4. Zeros -1, 0, 4; signs $-, +, -, +$; $-1 < x < 0$ or $x > 4$, i.e. $(-1, 0) \cup (4, \infty)$
5. Zeros -1, 5; signs $+, -, +$; $x \leq -1$ or $x \geq 5$, i.e. $(-\infty, -1] \cup [5, \infty)$
6. Zeros -3, 3; signs $+, -, +$; $x < -3$ or $x > 3$, i.e. $(-\infty, -3) \cup (3, \infty)$
7. Zeros -1, 4; signs $+, -, +$; $-1 < x < 4$, i.e. $(-1, 4)$
8. Zeros -2, 0, 2; signs $-, +, -, +$; $x < -2$ or $0 < x < 2$, i.e. $(-\infty, -2) \cup (0, 2)$
9. Zeros -1, 2; signs $-, +, +$; $x > -1$, $x \neq 2$, i.e. $(-1, 2) \cup (2, \infty)$
10. Zeros 2, 3; signs $+, -, +$; $2 \leq x \leq 3$, i.e. $[2, 3]$
11. Zeros -2, 0, 3; signs $-, +, -, +$; $x < -2$ or $0 < x < 3$, i.e. $(-\infty, -2) \cup (0, 3)$
12. Zeros -3, 2; signs $+, -, +$; $x \leq -3$ or $x \geq 2$, i.e. $(-\infty, -3] \cup [2, \infty)$
13. Zeros -1, 0, 1, 2; signs $+, -, +, -, +$; $x < -1$ or $0 < x < 1$ or $x > 2$, i.e. $(-\infty, -1) \cup (0, 1) \cup (2, \infty)$